

3 Yr. Degree/4 Yr. Honours 2nd Semester Examination, 2026 (CCFUP)

Subject : Chemistry

Course: CHEM2011

(Basic Chemistry-II)

Time: 2 Hours

Full Marks: 40

*The figures in the right hand margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*

1. Answer any five questions:

2×5=10

- (a) What do you mean by the term 'nucleofugality'?
- (b) Balance the chemical reaction by ion-electron method:

$$\text{Zn} + \text{NO}_3^- + \text{OH}^- \longrightarrow \text{ZnO}_2^{2-} + \text{NH}_3$$
- (c) Prove that $dS = C_p dT/T - \alpha V dP$.
- (d) Define μ bond. Give an example.
- (e) Write down the Maxwell velocity distribution equation for 1D, 2D and 3D.
- (f) *Cis*-1,2-dichloroethene boils at a higher temperature but melts at a lower temperature than the *trans* isomer.—Explain.
- (g) Find out the oxidation states of the underlined elements given below:
 Ba_2XeO_6 , $\text{Na}_2\text{S}_2\text{O}_3$, $\text{H}_2\text{S}_2\text{O}_8$ and CrO_5
- (h) (R) Phenyl sec-butyl ketone undergoes racemization with aqueous NaOH solution. — Explain.

2. Answer any two questions:

5×2=10

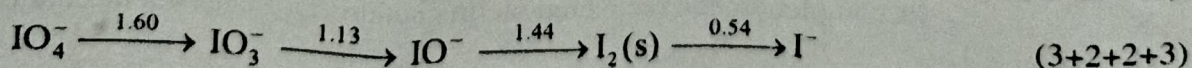
- (a) (i) Find ΔG_{mix} and ΔV_{mix} when 1 mole of benzene is mixed with 2 moles of toluene at 20°C and 1 atm. The solution is ideal. Which of the given data is redundant?
- (ii) 1 mole of an ideal gas undergoes a change from $(P, V, T) \longrightarrow (P/m, mv, T)$. Show that $\Delta S = R \ln m$. 3+2
- (b) (i) Comment on enol content of $(t\text{BuCO})_3\text{CH}$.
- (ii) In the reaction of $\text{Cl}_2\text{C}=\text{CHCl}$ with NaOD/D₂O, when the reaction is stopped before its completion, the recovered olefin contains D. Suggest a mechanism of the reaction. 3+2

- (c) (i) XeO_6^{4-} smoothly oxidises Mn (II) in acid medium, itself being reduced to XeO_3 -. Balance the reaction by ion-electron method.
- (ii) The electrode potential values of $\text{Cu}^{2+}/\text{Cu}^+$ and $\frac{1}{2} \text{I}_2/\text{I}^-$ suggest that iodometric titration of Cu^{2+} ion is not possible, but in practice it is found otherwise. — Explain. 3+2
- (d) (i) Give the mechanism of oxidation of isopropyl alcohol with chromium (IV) oxide in glacial acetic acid. Indicate whether 1° KIE is operative here and cite a labelling experiment in favour of your answer.
- (ii) An optically pure sample of R(-)-2-butanol shows a specific rotation of (-) 13.6° . What relative molar proportion of S (+)-2-butanol and R (-)-2-butanol would give a specific rotation of 6.8° ? (1.5+1.5)+2

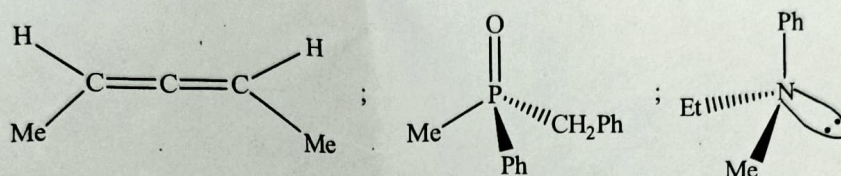
3. Answer any two questions:

10×2=20

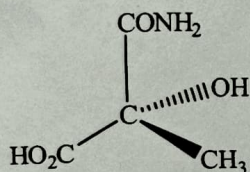
- (a) (i) Plot the velocity distribution curve for 3D, showing most probable velocity (C_{mp}), root mean square velocity (C_{rms}) and average velocity (C_{av}) in the curve for an ideal gas at a definite temperature.
- (ii) A mercury droplet of diameter 3 mm breaks into 8 identical smaller droplets. Calculate the change in surface energy. [Surface tension of Hg = 490 dyne cm^{-1}].
- (iii) Prove that $C_p - C_v = \alpha^2 TV/\beta$, where C_p = heat capacity at constant pressure, C_v = heat capacity at constant volume, α = co-efficient of thermal expansion, β = co-efficient of isothermal compressibility.
- (iv) Gases can't be liquefied unless their temperature is $\leq T_c$ — is it correct? (2.5+2.5+3+2)
- (b) (i) Account for the following change of 1-alkene / 2-alkene ratio as the base is changed in the dehydrobromination of 2-bromo-2,3-dimethylbutane:
- | | | | | |
|-------------------|---|----------------|--------------------------|--------------------------|
| Base | : | EtO^- | Me_3CO^- | Et_3CO^- |
| 1-alkene/2-alkene | : | 0.25 | 2.7 | 11.4 |
- (ii) What happens when (R)- α -bromopropionic acid is treated with NaOH solution?
- (iii) What is reflection invariant? Give one example. 4+3+3
- (c) (i) Describe the hybridization in case of PCl_5 . Why are the axial bonds longer as compared to equatorial bonds?
- (ii) No pseudorotation takes place in PCl_3F_2 molecule. — Explain.
- (iii) Explain resonance energy in terms of wave mechanics.
- (iv) Calculate E^0 for the couple $\text{IO}_3^- - \text{I}^-$ from the Latimer diagram (acid medium; E^0 in Volt):



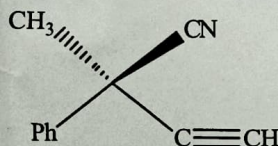
- (d) (i) The addition of Cl_2 (1 mol) to 1,3-butadiene at 25°C gives 60% 3,4-dichloro-1-butene and 40% 1,4-dichloro-2-butene. At 200°C , the yields are 30% and 70% respectively. Explain these observation with an energy profile diagram.
- (ii) Calculate the lattice energy of KCl. [$A = 1.746$, $r_0 = 3.08 \text{ \AA}$, $\epsilon_0 = 8.85 \times 10^{-12}$, $n = 9$].
- (iii) Explain whether the following compounds are resolvable or not:



- (iv) Give R/S configurational descriptor of the following compounds. Mention the priority sequence of the ligands around the chiral centre.



(I)



(II)

3+2+3+2